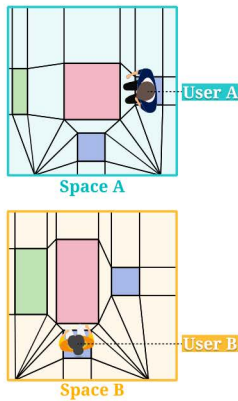


Spatial Affordance-Aware Affine Transformation Between Heterogeneous Spaces for Mixed Reality Remote Collaboration

Seonji Kim , Dooyoung Kim , Selin Choi , and Woontack Woo 

(a) Heterogeneous Spaces



(b) Spatial Affordance-Aware Collaboration

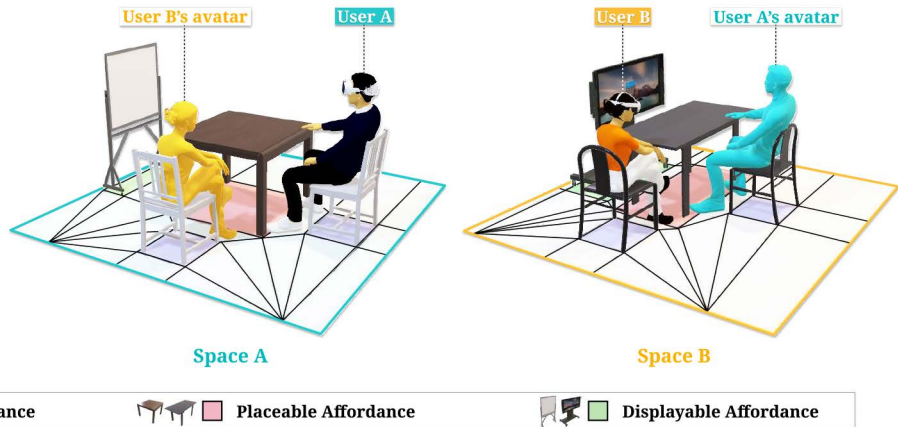


Fig. 1: This figure illustrates a mixed reality remote collaboration scenario after applying the affine patch-based transformation. (a) Top-down views of heterogeneous Space A and Space B annotated with spatial affordance-aware affine patches. Avatar motion across spaces follows the scale ratios between matched patches. (b) Following user movement across matched patches, avatars move to corresponding locations and sit on mapped chairs, forming an L-shaped collaborative arrangement consistent with spatial affordances.

Abstract—We propose a spatial affordance-aware affine transformation method between heterogeneous spaces for continuous multi-object matching in shared Mixed Reality (MR) spaces. While previous redirection and spatial mapping approaches utilize physical objects and walkable areas, a critical gap remains in enabling continuous mapping between dissimilar physical environments that supports both precise object alignment and seamless locomotion in a shared space. Our method structurally segments heterogeneous spaces into interaction zones and constructs affine patches based on object adjacency and facing configuration, enabling continuous correspondence. We evaluate our method using a dataset of paired dissimilar spaces and demonstrate that, unlike conventional grid-based methods, our approach achieves broader spatial alignment and richer object matching. The results show that our method can serve as an effective mapping framework for shared environments requiring semantic continuity and structural coherence across diverse real-world spaces.

Index Terms—Virtual Reality, Augmented Reality, Mixed Reality, Spatial Affordance, Heterogeneous Spaces, Space Mapping, Affine Transformation, Remote Collaboration

1 INTRODUCTION

Remote collaboration using Augmented Reality and Virtual Reality (AR/VR) head-mounted displays (HMDs) offers significant value by enabling users to communicate and collaborate beyond physical constraints [31, 55]. In particular, remote participants can appear as 3D avatars in the host's environment through AR/VR HMDs [39, 52]. However, the spatial layout of the virtual collaborative space often differs from each user's physical surroundings. To support walkable continuity and object-centric interaction, it is essential to reconcile these discrepancies by establishing affordance-relevant geometric and

semantic alignment [17]. With such alignment, users can navigate and manipulate without physical collisions or disjointed interactions. Remapping and redirection techniques can complement this alignment by managing locomotion and space reuse [23]. Therefore, addressing heterogeneous physical environments with an affordance-aware spatial alignment grounded in both geometry and semantics is critical for reliable remote collaboration.

Attempts have been made to reconcile the movable areas of dissimilar physical environments by intersecting them to generate semantically aligned mutual space [17, 25, 43]. However, these methods rely on geometric matching that overlooks user-centered spatial affordances, and often result in a limited locomotion with narrower shared space. Another line of research on object-level spatial affordance has either focused on detailed interactions with individual objects [8, 9] or attempted to support complex interactions involving multiple objects [24], but it had limitations in capturing both the multiplicity and precision of object interactions simultaneously. There was an approach that aligned shared experiences based on spatial affordances and key elements; however, limited in that only a subset of the physical space can be effectively utilized for collaboration [20]. Space transformation techniques have been proposed to address these limitations, feasible even in highly dissimilar

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space [3, 23, 51, 54]. However, they suffer from resets, discontinuities, and limited interactivity, often requiring continuous target prediction or manual selection, which poses practical challenges. Therefore, to enable realistic collaboration across dissimilar spaces, there is a clear need for a spatial mapping method that considers the affordances of multiple objects while also maximizing the continuity and extent of the shared virtual environment.

We present a spatial mapping method for remote collaboration that supports object-centric correspondence and walkable continuity across dissimilar environments. When the shared space is generated before collaboration, the system computes the mapping from pre-labeled scene inputs; during use the selected piecewise mapping is applied deterministically, with no mid-session re-optimization. Fig. 1 summarizes the method and its three goals (continuity, determinism, persistence), which promote spatial coherence and stable correspondences. To organize affordance-relevant geometry, we adopt a centrality-based zoning model inspired by proxemics (O-zone: center object; P-zone: surrounding objects and floor; R-zone: peripheral wall-facing areas) and prioritize mapping in the order $O \rightarrow P \rightarrow R$ so that task-relevant regions are preserved first. Building on this, our piecewise algorithm (Affine Patch) proceeds in four stages: object matching, affine patch generation, obstacle registration, and optimal map selection. Four patch types (center-facing, side-by-side, corner-adjacent, boundary-aligned) are instantiated from adjacency and facing relations, with patches primarily placed in P and R to maintain task surfaces and walkable continuity. Among candidate mappings derived from alternative object matchings, the final mapping is chosen by maximizing a weighted objective based on affordance-preserving criteria.

To evaluate the effectiveness of the proposed spatial affordance-aware affine patch mapping, we compared it with two grid-based baselines ($1 \times 1 \text{ Grid}$, $3 \times 3 \text{ Grid}$) on a dataset of 10 dissimilar space pairs. We examined three aspects: (i) registered walkable area between spaces, (ii) preservation of object-level affordances and edge alignment on task surfaces, and (iii) the magnitude of spatial deformation (ΔT). The method improved object-edge alignment over $3 \times 3 \text{ Grid}$ by $\approx 29\%$ (0.662 vs. 0.513; $p_{\text{adj}} = 0.001$). Relative to the coarse $1 \times 1 \text{ Grid}$, it expanded task-relevant regions: object-area mapping by $\approx 88\%$ (0.758 vs. 0.403) and walkable-area mapping by $\approx 1.6\%$ (0.992 vs. 0.976), while remaining statistically comparable to $3 \times 3 \text{ Grid}$ on these area measures (OP: 0.758 vs. 0.744, $p_{\text{adj}} = 1.000$; WA: 0.992 vs. 0.991, $p_{\text{adj}} = 0.876$). For deformation cost, no significant differences were detected across methods (Friedman $\chi^2(2) = 2.60$, $p = .273$; $3 \times 3 \text{ Grid}$ vs. *Affine*: $p_{\text{adj}} = 0.393$), indicating similar cost in our setting.

Our contributions are threefold. First, we propose a spatial affordance-aware space transformation method, spatial affordance-aware affine patch mapping, that preserves continuous, affordance-relevant correspondences across heterogeneous physical environments by instantiating three design principles: affine equivalence, affordance-aware decomposition, and deterministic correspondence mapping. Second, comparative evaluations on dissimilar pre-labeled, static scene pairs show that our method registers broader continuous task-relevant regions (placeable and walkable areas) than coarse grids and supports a wider range of object-level interactions, while maintaining deformation cost (ΔT) statistically comparable to a fine grid. Third, we demonstrate robustness as a cost-neutral improvement, with stable correspondences under variations in layout and affordance configurations, by computing the mapping pre-session and deploying it deterministically at runtime. These results highlight the method's practical applicability to real-world MR collaboration systems and advance the spatial-mapping capabilities essential for immersive, intuitive AR/VR collaboration. Looking ahead, the piecewise-affine design provides a scalable foundation for extensions to multi-cluster and dynamic environments, supporting future applications in remote communication, education, and cultural sharing.

2 RELATED WORKS

2.1 Shared Virtual Space Between Physical Rooms

A virtual shared space where users meet and interact using AR/VR HMDs is an essential element for remote collaboration. A virtual shared space that allows homogeneous interaction for all participants in

given digital twin spaces is essential for life-like remote collaboration and is generated based on common space elements between different dissimilar spaces [43]. Some studies have created shared spaces with common floors or walls, but the objects inside were considered only as backgrounds [25, 30]. In order to create a sense of reality by constructing a virtual shared space rooted in the physical environment, physical objects in the space must be interactable, but this was not considered in these studies [12]. There have been methods that approach objects as obstacles to walking when experiencing virtual spaces in physical spaces, but in these cases, the focus is on the geometric information of the objects, making interaction with the objects still impossible [51].

To address these limitations, it is crucial to consider spatial affordances—the unique interaction opportunities defined by a space's geometric, functional, and semantic properties. In this study, we aim to identify common spatial elements between two spaces by leveraging the inherent attributes of individual objects, as demonstrated in previous research on scene synthesis [26, 29, 35, 56] and virtual object or interface placement [2, 6, 36]. Unlike earlier mutual scene generation studies that additionally use such information to extend perceptual collaboration space [17, 18, 32], we propose to comprehensively integrate spatial affordance properties to maximize an interactable shared virtual space rooted in the physical environment. This allows us to generate an interactable shared virtual space between dissimilar rooms, including empty floors and interior objects.

2.2 Physical Object-centric Remote Collaboration

Herskovitz et al. [11] named the term ‘Object-Centric Methods’ to refer to distributed AR/VR collaboration utilizing center physical objects as anchors or references. Room2Room [34] was one of the object-centric methods in the research domain, and Spatial¹ or Mesh² were introduced as commercial applications. Many object-centric collaboration methods have been studied targeting specific single interaction objects such as desks, monitors, and whiteboards according to collaboration scenarios, but most of them did not consider the difficulty of interaction due to geometric dissimilarity between target objects [4, 9, 40]. To mitigate dissimilarities between objects, research has proposed methods to enhance single object-level interaction quality through object boundary alignment of a central object [23]. However, these approaches fail to consider the different types and numbers of interacting objects, and as a result, complex multi-object spatial affordances are not adequately considered.

In real-world environments, spatial affordance often involves multiple objects simultaneously, such as using a ‘desk’ while sitting in a ‘chair’ or moving between a ‘chair’ and a ‘whiteboard’. Although face-to-face collaboration naturally accommodates these complex interactions, current remote collaboration methods are limited. To address this, dynamic space-matching techniques that alternate between horizontal and vertical interaction objects have been proposed [8], but their applicability to real environments remains unaddressed. There have been attempts to address this by using scene graphs as a data structure to represent physical environments that enable complex interactions [50]. A scene graph is a comprehensive data structure that represents all elements of a scene as nodes and the relationships between them as edges [1, 13, 27, 41, 45]. Specifically, recent research has explored using scene graphs to find target objects or to form clusters of co-interacting objects and create shared spaces reflecting these interactions [20, 24]. However, these methods tend to consider subsets of objects or spaces within the aligned clusters, limiting the ability to capture precise multi-object interactions. Therefore, this study enhances object-level interaction by broadly considering affordance-driven interactions between central and surrounding objects, while also enabling precise alignment at the level of individual object edges.

2.3 Space Transformation for Room Dissimilarity

The inherent diversity and complexity of real environments make it difficult for intersection-based methods to effectively account for spatial

¹<https://www.spatial.io/>

²<https://www.microsoft.com/en-us/microsoft-teams/microsoft-mesh>

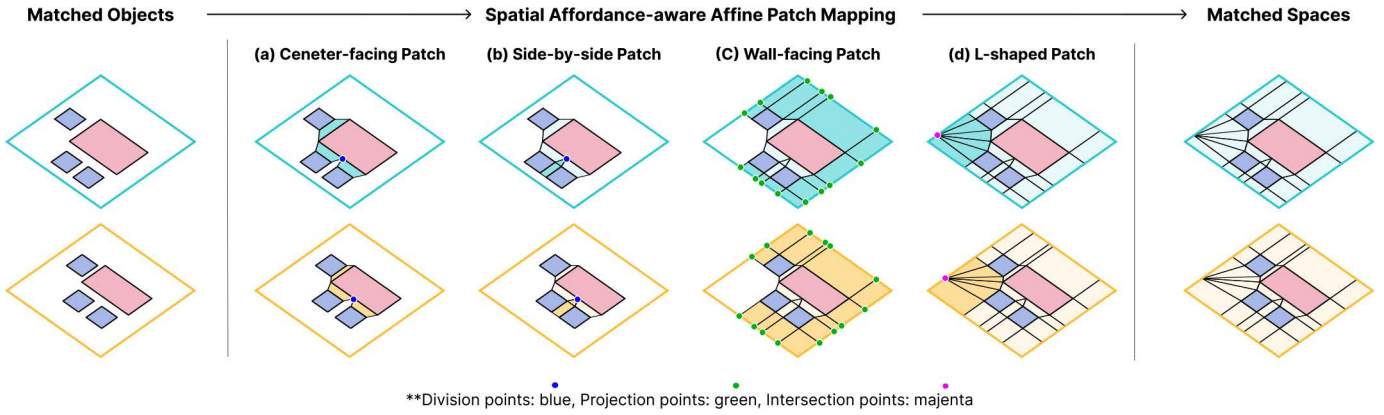


Fig. 2: Four types of affine patch generation: (a) center-facing, (b) side-by-side, (c) wall-facing, and (d) L-shaped. These patches are generated sequentially to segment the walkable space into affine regions. The red areas indicate the affine patches mapped at each stage.

Table 1: Summary of patch types based on facing formation, used geometric elements, and resulting shape.

Patch Type	Facing Formation	Key Elements	Patch Shape
Center-facing Patch	Center object-facing	Division points subdividing center edge	Trapezoid
Side-by-side Patch	Side-by-side	Division points subdividing center edge	Trapezoid + triangle
Wall-facing Patch	Wall-facing	Projection points on wall	Trapezoid or triangle
L-shaped Patch	L-shaped	Intersection point from angle bisector	Triangles (×4) with residual regions

affordances in shared virtual spaces, including locomotion and object interaction. To overcome this, various space transformation techniques have been explored to support interaction from spatial affordances, even across highly mismatched environments [49]. Methods to reposition the remote user's avatar and adjust its pose have been studied, but they do not account for the continuity of interaction, often causing distortion and reset that hinder immersion and collaboration realism [7, 15, 46, 53, 54]. In addition, there have been studies utilizing the Redirected Walking (RDW) technology, which is a method of manipulating the user's perceived movement direction [37]. However, RDW also has limitations in that distortion and reset occur when the threshold is exceeded [5, 38, 44, 47, 48]. Therefore, a space transformation method is needed that enables corresponding regions to be aligned without redundancy or discontinuity, in order to overcome spatial dissimilarity.

Space scaling was studied as a space transformation method without reset even based on RDW technology, and it was applied to the entire space, mainly focusing on empty area [19, 21, 22]. In addition, there was a study that focused on securing the interaction of the central object by creating a 3×3 grid based on its edge (named boundary in this study) and applying space rescaling to each cell [23]. However, the rescaling method had limitations due to the rescaling threshold from RDW and limited spatial affordance consideration. In addition, there was a method that mapped the position of the avatar based on the same points between spaces [3]. However, the process of extracting matching points in space was not clarified, and objects were considered as obstacles, so various interactions were not considered. Therefore, this study proposes a new space transformation method, piecewise space mapping, that overcomes the limitations of the methods. The proposed method is based on a spatial affordance-aware affine patch without duplication, enabling continuous and stable interactions even among highly dissimilar remote spaces.

3 METHODOLOGY

3.1 Design Principle

We propose a spatial mapping technique that enables remote users to interact with objects and navigate the environment as if they were physically co-present in the local space. Our method is built upon three core goals: *Continuity*, *Determinism*, and *Persistence*. These principles guide how we structure spatial affordances, ensure reliable

interaction, and provide a seamless user experience, to map between dissimilar environments.

- *Continuity* ensures that the user's avatar can move and interact within the environment without interruption. By maintaining spatial coherence, the method supports consistent navigation and interaction.
- *Deterministic mapping* guarantees that once a correspondence between spaces is defined, it remains fixed. Users can rely on stable spatial relationships without repeated computation.
- *Persistence* maintains the same mapping throughout an interaction. As long as the interaction context continues, transformation differences are minimized, and the flow of interaction remains stable.

To satisfy these goals, we propose a spatial affordance-aware piecewise mapping method that establishes correspondences between two spaces and enables dissimilar space mapping. A *affine patch* is a convex polygonal region that supports affine-equivalent transformation and is defined based on spatial elements that determine the spatial affordance of a given environment. Following the design principles below, we use these patches to construct a piecewise mapping that achieves a continuous, deterministic, and persistent inter-space correspondence.

1. An affine patch must be unique and correspond one-to-one without redundancy. To guarantee continuity, patches in different spaces should be connected by equivalent sets of points and edges, ensuring a seamless geometric relationship.
2. During the construction of each affine patch, unmatched objects or regions are treated as virtual obstacles. These are augmented using an affine patch-based transformation to maintain consistent correspondence across the mapped space.
3. Each affine patch reflects spatial affordances and includes all elements relevant to interaction. Segmentation leverages Kendon's F-formation and interaction space, actively using object and wall edges to support interaction-driven partitioning [16].

When dissimilar physical spaces are given as input data, the space transformation is completed through four processes: 1) object mapping, 2) affine patch mapping, 3) obstacle augmentation, and 4) optimal map

selection. Our method assumes collaborative scenes centered around a key object, following prior studies on structured collaboration around central objects [20, 23]. Based on our third design principle, we apply F-formation to room-scale clusters, interpreting the area around the center object as the *O-zone* (corresponding to O-space), the area containing surrounding objects as the *P-zone* (P-space), and the area outside the cluster as the *R-zone* (R-space). Mapping is prioritized in order of spatial centrality: the center object in the O-zone is mapped first, followed by surrounding objects and floor regions in the P-zone, and then wall-facing open spaces in the R-zone. In the P-zone, floor regions not occupied by surrounding objects are mapped using affine patches, categorized into four types based on object facing-formation. In the R-zone, the empty space between walls and objects is mapped using affine patches. Unmatched obstacles in remote areas are transformed via their corresponding affine patches and allocated as virtual obstacles. Among all mapped space configurations generated through different object matching combinations, the one that best satisfies the objective function is selected as the final mapped space. The overall process is illustrated in Fig. 2, and details of each process are described in the following section.

3.2 Procedure

3.2.1 Object Mapping

Our method operates on the top-down view of a space, assuming a preprocessed scene representation in which objects are identified with semantic affordances and geometric properties, including bounding boxes. In this work, we focus on three object affordances that are highly relevant to remote collaboration scenarios—placeable, sittable, and displayable—as identified in prior work [24, 33, 42]. In each space, the object with the placeable property that has the greatest potential for facing other objects is treated as the center object [24].

According to the design principle, the center object of the O-zone is allocated first, and then the surrounding objects of the P-zone are allocated. Surrounding objects are matched sequentially based on four facing directions (top, bottom, left, and right) from the center object to account for object-level spatial affordances. This process generates all valid matching conditions between the two center-facing configurations. For each direction, it computes all feasible pairwise matchings between facing objects and combines them via the Cartesian Product to enumerate all consistent combinations. Objects that do not match each condition are classified as obstacles and will be processed in the obstacle augmentation phase. To ensure topological consistency, the clockwise ordering of matched objects around the center must be preserved, avoiding any twisted or flipped configurations.

3.2.2 Affine Patch Mapping

Fig. 2 shows the second process, affine patch mapping, which partitions the floor area with affine patches considering the matched objects from the first step. In this stage, affine patch mapping is performed across the P-zone and R-zone. The core idea of this stage is to identify the floor area between adjacent objects and fill it with an affine-equivalent polygon to form an affine patch. Adjacent refers to the inter-object condition where no other object is located between the two. Our method builds upon prior work that applied the concept of facing formation to object clusters to determine which edges or corner points of objects are used in patch generation [24]. Emphasizing direct and continuous interaction between adjacent objects, our approach ensures that the generated patches remain unique and geometrically continuous under affine transformation. Depending on the spatial elements involved in patch construction, four types of patches are defined: Center-facing Patch, Side-by-Side Patch, L-shaped Patch, and Wall-facing Patch, as summarized in Tab. 1. Detailed construction methods for each patch type are described in the following sections.

Center-facing Patch is created when a surrounding object is adjacent to and directly facing the center object, with no object in between. The patch uses the facing edges of both objects. As shown in Fig. 2(a), if there is only one surrounding object on the side, a trapezoidal patch is formed by connecting the endpoints of both edges. When multiple

facing objects compete for the same boundary region, to prevent conflicts around the center object, division points are placed at the center of the gaps between adjacent objects and projected onto the center edge. Each resulting segment is used to form a separate affine patch with the corresponding facing object.

Side-by-Side Patch is created when multiple surrounding objects are adjacent to each other along the same side of the center object without facing it, forming a side-by-side facing formation. In the side-by-side formation, determining division points is key to preventing boundary conflicts between adjacent furniture objects by explicitly subdividing the shared edge region, illustrated as blue dots in Fig. 2(b). Division points, computed as in the center-facing patch, are projected onto the center object's edge to subdivide the boundary. Each object forms two affine patches: one trapezoidal and one triangular, using the corresponding division point and its own adjacent edge.

Wall-facing Patch is created when a surrounding object is adjacent to and facing a wall, with no object in between. As shown in Fig. 2(c), the patch is defined using the endpoints of the object's facing edge. These endpoints are projected toward the wall to generate projection points (green dots in Fig. 2(c)), which determine the patch boundary. The projection direction is determined by the object's position relative to the wall and the orientation of its facing edge.

L-shaped Patch is created when two surrounding objects are adjacent to perpendicular edges around a corner of the center object, forming an L-shaped facing formation. As shown in Fig. 2(d), the patch uses the corner point of the center object and the four endpoints of the adjacent surrounding edges. Among the angles formed by these points, the smallest is bisected, and the bisector is extended toward the wall to determine an intersection point. This point is then connected with all other relevant points to divide the complex corner region without overlap. This decomposition yields four primary triangular patches, while any remaining regions around the corner are naturally partitioned into triangular or quadrilateral patches.

3.2.3 Obstacle Augmentation

The third stage is obstacle augmentation, which is performed after all affine patches have been generated. In this stage, we handle the remaining objects in the remote space that were previously classified as obstacles and excluded from the mapping process. Since these obstacles do not exist in the local space, they can interfere with the user's locomotion or interaction by causing unintended collisions with the avatar. To prevent this, we locate the corresponding region in the local space and insert a virtual obstacle to represent the remote obstacle, ensuring that the walking experience is not disrupted. This is achieved by identifying the affine patch in the remote space that intersects with the physical obstacle and finding its corresponding patch in the local space. Because each pair of matched affine patches shares a one-to-one correspondence between all corner points, the corresponding subregions of obstacle objects can be reliably transformed. An affine transform is then applied to transfer the obstacle to the local space. If the obstacle already overlaps with a patch that has been marked as an obstacle, no virtual obstacle is created. When a physical obstacle intersects with multiple affine patches, the obstacle is split according to the patch boundaries, and the corresponding transform is applied to each part individually.

3.2.4 Optimal Map Selection

$$O_{\text{map}} = \omega_1 M_{\text{obj}}^{\text{area}} + \omega_2 M_{\text{obj}}^{\text{edge}} + \omega_3 M_{\text{walk}} \quad (1)$$

The final stage identifies the best mapping result among all spatial mapping candidates. The optimal condition is selected based on the objective function shown in Eq. (1). This function quantifies the degree to which spatial affordances are preserved in the mapped space. The objective function consists of three weighted terms: the *object plane mapping rate*, the *object edge mapping rate*, and the *walkable area mapping rate*. The object-related terms assess how well object-level spatial affordances are preserved, while the walkable area term evaluates the preservation of floor-level affordances for locomotion and navigation.

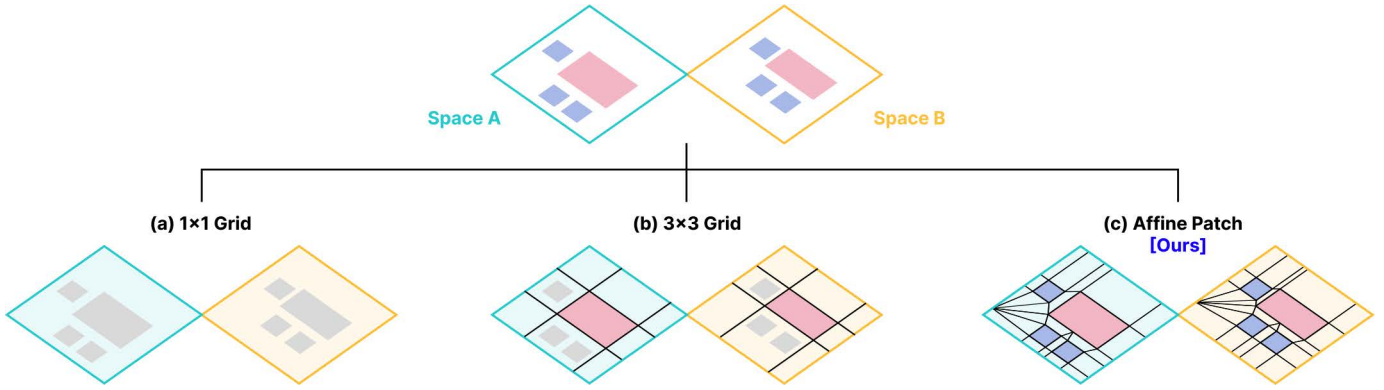


Fig. 3: This figure illustrates our method (*Affine Patch*) and two baselines (*1×1Grid*, *3×3Grid*). (a,b) *1×1Grid* maps the room as a single patch using wall geometry, while *3×3Grid* divides it into 9 patches based on walls and a central object. (c) In contrast, *Affine Patch* maps the room, central object, and surrounding objects into triangular or convex patches, incorporating spatial affordances.

Here, ω_1 , ω_2 , and ω_3 are weight coefficients for each term. $M_{\text{obj}}^{\text{area}}$ and $M_{\text{obj}}^{\text{edge}}$ represent the object plane and edge mapping rates, respectively, and quantify how effectively object geometries are preserved in the mapping. M_{walk} measures the proportion of navigable space successfully mapped to maintain spatial continuity for user movement. Each term is computed based on the portion of spatial elements—area or edge length—that are successfully mapped in each physical space. The detailed computation methods are described as follows.

Object Plane Mapping Rate ($M_{\text{obj}}^{\text{area}}$) measures the average proportion of object surface area that is included in valid mappings across multiple spaces:

$$M_{\text{obj}}^{\text{area}} = \frac{1}{N} \sum_{i=1}^N \frac{A_{\text{obj},i}^{\text{map}}}{A_{\text{obj},i}} \quad (2)$$

where $A_{\text{obj},i}$ denotes the total surface area of all objects in space i , and $A_{\text{obj},i}^{\text{map}}$ is the portion that is successfully mapped.

Object Edge Mapping Rate ($M_{\text{obj}}^{\text{edge}}$) measures the average proportion of object edge length preserved in the mapping:

$$M_{\text{obj}}^{\text{edge}} = \frac{1}{N} \sum_{i=1}^N \frac{E_{\text{obj},i}^{\text{map}}}{E_{\text{obj},i}} \quad (3)$$

where $E_{\text{obj},i}$ is the total edge length of all objects in space i , and $E_{\text{obj},i}^{\text{map}}$ is the portion included in the mapping.

Walkable Area Mapping Rate (M_{walk}) evaluates the average proportion of navigable space that is successfully mapped:

$$M_{\text{walk}} = \frac{1}{N} \sum_{i=1}^N \frac{A_{\text{walk},i}^{\text{map}}}{A_{\text{walk},i}} \quad (4)$$

where $A_{\text{walk},i}$ is the total walkable area in space i , and $A_{\text{walk},i}^{\text{map}}$ is the mapped portion. In this paper, we evaluate mappings between two dissimilar spaces, and thus use $N = 2$ for all the above equations.

4 EVALUATION

4.1 Study Design and Implementation

We evaluate *Spatial Affordance-aware Affine Patch Mapping* (hereafter *Affine Patch*) with two grid-based baselines adapted from prior work—*1×1Grid* [22] and *3×3Grid* [23]. The key differences among the three methods lie in the number and shape of patches (dashed and solid lines) and in the spatial information considered during patch generation; objects *not* considered by a given method are shown in gray, as illustrated in Fig. 3. As shown in Fig. 3(a), *1×1Grid* treats the entire floor area as a single patch; object information within the patch is transformed passively and is not explicitly used for patch generation.

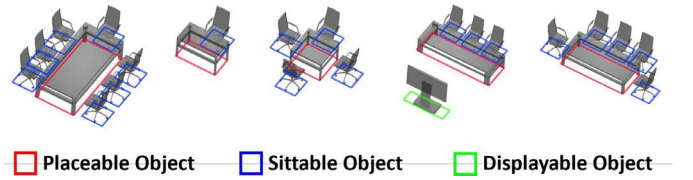


Fig. 4: This figure illustrates sample object clusters from real-world environments used to construct dissimilar space pairs in our experiments, reflecting furniture configurations commonly seen in offices and rooms.

In contrast, *3×3Grid* (see Fig. 3(b)) actively incorporates the central object by extending its edges to divide the floor into nine patches in a 3×3 grid. The proposed *Affine Patch* method, illustrated in Fig. 3(c), considers both the central and surrounding objects to construct affine-equivalent polygons that fill the floor area *without overlaps or gaps*. We assess spatial affordance preservation using three outcome measures—registered walkable area, object-area mapping on task surfaces, and object-edge alignment—and additionally measure the *Transform Magnitude* (ΔT) to quantify deformation cost.

Our implementation runs on a desktop with an Intel Core i9-13900KF CPU, NVIDIA RTX 4090 GPU, 32GB RAM, and 4 TB storage. We utilize standard open-source Python libraries, including NumPy (v2.0.1) and Pandas (v2.2.2) for data handling [10, 28], Shapely³ (v2.0.6) for geometric computations, Matplotlib (v3.9.1) for visualization [14], and Pillow⁴ (v10.4.0) for image processing. To ensure a controlled and consistent evaluation across methods, our approach assumes a single central placeable object in each mapping scenario, in alignment with the algorithm's intended design. For the experiment, all coefficients in Eq. (1), ω_1 , ω_2 , and ω_3 , were equally weighted as $\frac{1}{3}$.

4.2 Space Pair Dataset

As described in Sec. 4.1, we compared our method with two grid-based baselines (*1×1Grid*, *3×3Grid*). To enhance practical applicability, the spatial datasets were constructed based on real-world spaces (floor plans and object clusters) under predefined conditions of spatial dissimilarity, while object affordances and geometric properties were explicitly specified for controlled evaluation. Because our goal is to generate shared spaces for remote collaboration, we used floor plans from common meeting rooms, offices, and living rooms that are typically used by multiple people. These plans were created with RoomSketcher⁵. The base floor area was taken from an actual multi-cluster environ-

³Documentation: <https://shapely.readthedocs.io>

⁴Documentation: <https://pillow.readthedocs.io>

⁵<https://www.roomsketcher.com>

Table 2: Mapping scores (mean M , standard deviation SD) and statistical test results across methods. Post-hoc comparisons were Bonferroni-corrected across the four outcome families. For non-normal outcomes (ΔT), medians and interquartile ranges (IQR) are also reported in the text/appendix to reflect distributional skew. (Significance legend: ns = not significant ($p > .05$), * = $p \leq .05$, ** = $p \leq .01$, *** = $p \leq .001$)

Metric	Method	M	SD	Test Result
Object Area Mapping Rate	$1 \times 1 \text{Grid}$	0.403	0.179	ANOVA: $F(2, 18) = 39.75$ ***
	$3 \times 3 \text{Grid}$	0.744	0.057	
	$Affine\ Patch$	0.758	0.072	
Post-hoc: 1×1 vs 3×3 (***), 1×1 vs Affine (***), 3×3 vs Affine (ns)				
Object Edge Mapping Rate	$1 \times 1 \text{Grid}$	0.128	0.149	ANOVA: $F(2, 18) = 74.07$ ***
	$3 \times 3 \text{Grid}$	0.513	0.045	
	$Affine\ Patch$	0.662	0.084	
Post-hoc: 1×1 vs 3×3 (***), 1×1 vs Affine (***), 3×3 vs Affine (**)				
Walkable Area Mapping Rate	$1 \times 1 \text{Grid}$	0.976	0.005	ANOVA: $F(2, 18) = 57.37$ ***
	$3 \times 3 \text{Grid}$	0.991	0.004	
	$Affine\ Patch$	0.992	0.004	
Post-hoc: 1×1 vs 3×3 (***), 1×1 vs Affine (***), 3×3 vs Affine (ns)				
Transform Deformation Cost	$1 \times 1 \text{Grid}$	2.489	3.213	Friedman: $\chi^2(2) = 2.60$ ns
	$3 \times 3 \text{Grid}$	1.869	1.114	
	$Affine\ Patch$	2.847	1.549	
Post-hoc: 1×1 vs 3×3 (ns), 1×1 vs Affine (ns), 3×3 vs Affine (ns)				

ment containing two or more single clusters and measures 121.27 m^2 ($13.43 \text{ m} \times 9.03 \text{ m}$). All generated space datasets share this base floor area so that observed effects arise from differences in object clusters. Each object cluster includes at least two of three interaction affordances: sittable, displayable, and placeable.

We conducted the evaluation on 10 distinct single-cluster pairs formed by pairing the space datasets shown in Fig. 4. The pairs exhibit substantial differences in spatial affordance due to variations in object types, counts, and layouts. The furniture sets reflect everyday office and home configurations (e.g., desks, chairs, displays, sofas, shelves), keeping scenarios representative while covering heterogeneous arrangements. All data were generated using Unity 2021.3.51f1.

4.3 Quantitative Metrics

We evaluate how well spatial affordance is preserved under each mapping method by referencing the three metrics used in the objective function: the object-area mapping rate, the object-edge mapping rate, and the walkable-area mapping rate. Each metric is computed as defined in Eq. (1), measuring the proportion of affordances that remain available under the mapping. For object areas, we use both the object-area and object-edge rates, following prior work that emphasizes edge-level interactions in addition to planar alignment [23]; a location is counted only when semantic labels match and geometric alignment falls within a tolerance.

Unlike intersection-based alignment methods that assess affordance via overlap or intersection-over-union, our mapping-based evaluation measures how much of each space's affordance becomes available when mapped onto the other space [23, 24]. Because the mapping aims to preserve each space's internal structure, we evaluate affordance in the source space's own coordinate frame after applying the mapping. In addition, we assess the overall result in a unified top-down view, rather than separating horizontal and vertical cases as in prior work [23].

To evaluate the geometric stability of each patch-level mapping, we compute the *Transform Difference cost* (D_{patch}), which quantifies the discrepancy between the estimated affine transformation and an ideal (identity) transformation. The score consists of three terms: (1) the average residual error between the transformed and target points, (2) the deviation of the linear matrix from the identity matrix (capturing scale and shear changes), and (3) the magnitude of the translation vector. It is formally defined as:

$$D_{\text{patch}} = \frac{1}{N} \sum_{i=1}^N \left\| A \begin{bmatrix} \mathbf{p}_i \\ 1 \end{bmatrix} - \mathbf{q}_i \right\|_2 + \|A_{\text{linear}} - I\|_F + \|\mathbf{t}\|_2 \quad (5)$$

where $\mathbf{p}_i, \mathbf{q}_i \in \mathbf{R}^2$ are the i -th corresponding points in the source and target patches, respectively. $A \in \mathbf{R}^{2 \times 3}$ is the estimated affine transformation matrix, which is decomposed into a linear part $A_{\text{linear}} \in \mathbf{R}^{2 \times 2}$ and a translation vector $\mathbf{t} \in \mathbf{R}^2$. The term $\|\cdot\|_2$ denotes the Euclidean norm, and $\|\cdot\|_F$ denotes the Frobenius norm.

To summarize the distortion across the entire mapping, we compute both the mean and standard deviation of the patch-wise transform-difference score (D_{patch}). In the main analysis, we report transform magnitude ΔT (cm)—the mean centroid displacement between matched object pairs—while the patch-level score D_{patch} is provided in the appendix as a robustness check.

4.4 Results

1. Object Area Mapping Rate

The mean and standard deviation scores for the *Object Area Mapping Rate* across the 10 space pairs were: $1 \times 1 \text{Grid}$ ($M = 0.403$, $SD = 0.179$), $3 \times 3 \text{Grid}$ ($M = 0.744$, $SD = 0.057$), and *Affine Patch* ($M = 0.758$, $SD = 0.072$). A repeated-measures ANOVA revealed a significant main effect of mapping method, $F(2, 18) = 39.75$, $p < .001$. Bonferroni-corrected post hoc tests showed $1 \times 1 \text{Grid} < 3 \times 3 \text{Grid}$ ($p_{\text{adj}} < .001$) and $1 \times 1 \text{Grid} < \textit{Affine Patch}$ ($p_{\text{adj}} < .001$), with no reliable difference between $3 \times 3 \text{Grid}$ and *Affine Patch* ($p_{\text{adj}} = 1.000$).

2. Object Edge Mapping Rate

The mean and standard deviation values for the *Object Edge Mapping Rate* were: $1 \times 1 \text{Grid}$ ($M = 0.128$, $SD = 0.149$), $3 \times 3 \text{Grid}$ ($M = 0.513$, $SD = 0.045$), and *Affine Patch* ($M = 0.662$, $SD = 0.084$). A repeated-measures ANOVA indicated a significant main effect of method, $F(2, 18) = 74.07$, $p < 0.001$. Post-hoc pairwise t -tests (Bonferroni-corrected; reported as p_{adj}) confirmed differences across all pairs: $1 \times 1 \text{Grid} < 3 \times 3 \text{Grid}$ ($p_{\text{adj}} < 0.001$), $1 \times 1 \text{Grid} < \textit{Affine Patch}$ ($p_{\text{adj}} < 0.001$), and *Affine Patch* $> 3 \times 3 \text{Grid}$ ($p_{\text{adj}} = 0.001$).

3. Walkable Area Mapping Rate

The average scores and standard deviations for the *Walkable Area Mapping Rate* were: $1 \times 1 \text{Grid}$ ($M = 0.976$, $SD = 0.005$), $3 \times 3 \text{Grid}$ ($M = 0.991$, $SD = 0.004$), and *Affine Patch* ($M = 0.992$, $SD =$

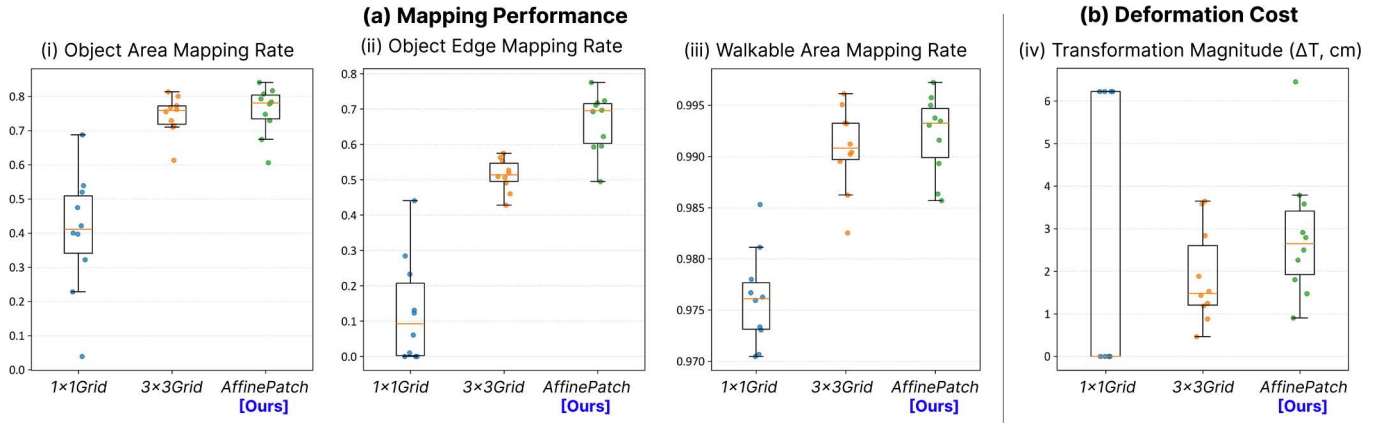


Fig. 5: Boxplots comparing mapping quality and deformation across methods. Left three panels show Mapping Performance Rates: (i) Object Area Mapping Rate (OP), (ii) Object Edge Mapping Rate (OE), and (iii) Walkable Area Mapping Rate (WA). The right panel shows Deformation Cost: Transform Magnitude (ΔT , cm). Boxes denote the interquartile range (IQR) with the median line; whiskers extend to $1.5 \times$ IQR; points indicate individual space pairs ($n = 10$). *Affine Patch* improves OE over 3×3 Grid (mean 0.662 vs. 0.513, $p_{\text{adj}} = 0.001$), is statistically comparable to 3×3 Grid on OP and WA (ns), and remains cost-comparable on ΔT (Friedman $\chi^2(2) = 2.60$, $p = .273$). See Table 2 for full statistics.

0.004). A repeated-measures ANOVA showed a significant main effect, $F(2, 18) = 57.37$, $p < .001$. Bonferroni-corrected post hoc tests indicated 1×1 Grid $< 3 \times 3$ Grid ($p_{\text{adj}} < .001$) and 1×1 Grid $< \text{Affine Patch}$ ($p_{\text{adj}} < .001$), with no reliable difference between 3×3 Grid and *Affine Patch* ($p_{\text{adj}} = 0.876$).

4. Transformation Difference

To evaluate how consistently corresponding object pairs were preserved during the mapping process, we analyzed the *mean difference* values provided in each method's output. Here, ΔT denotes the mean Euclidean displacement (cm) between matched object centroids after mapping, averaged over matched pairs. As shown in Tab. 2, the mean and standard deviation of this metric were as follows: 1×1 Grid ($M = 2.489$, $SD = 3.213$), 3×3 Grid ($M = 1.869$, $SD = 1.114$), and *Affine Patch* ($M = 2.847$, $SD = 1.549$). A Friedman test showed no significant effect across methods, $\chi^2(2) = 2.60$, $p = .273$. Bonferroni-corrected Wilcoxon tests also yielded no significant pairwise differences (1×1 vs 3×3 : $p_{\text{adj}} = 1.000$, 1×1 vs *Affine*: $p_{\text{adj}} = 1.000$, 3×3 vs *Affine*: $p_{\text{adj}} = 0.393$). Effect size by Kendall's W was 0.13.

5 DISCUSSION

5.1 Analysis

The proposed piecewise space transformation method demonstrated a clear advantage in mapping broader walkable areas between dissimilar environments. As shown in Fig. 5, the *Affine Patch* method reached near-ceiling walkable-area mapping and was comparable to 3×3 Grid (about 0.993 vs. 0.991), while clearly exceeding 1×1 Grid (about 0.978). Maintaining wide and continuous walkable regions is particularly important because it reduces interruptions and rerouting pressure during co-presence, making route continuity more robust even when two physical spaces are highly heterogeneous.

Our approach also improved spatial affordances at the object level, particularly in edge alignment. Relative to 3×3 Grid, the *Affine Patch* method increased the object-edge mapping rate by roughly +31% (≈ 0.651 vs. 0.496) while keeping object-plane mapping comparable (≈ 0.742 vs. 0.733) and still much higher than 1×1 Grid (≈ 0.386). By anchoring around the main task object and instantiating local affine patches in adjacent, facing, corner-adjacent, and boundary-aligned configurations, the method preserves adjacency and facing relations that matter for interactions. As a result, finer edge alignment expands the set of precise, edge-dependent interactions (e.g., tabletop boundaries and device rims) across multiple objects—not only the central one—without sacrificing overall planar coverage.

Although the analysis of transform magnitude (ΔT) did not reveal statistically significant differences across methods, the *Affine Patch*

mapping remained cost-neutral with respect to deformation compared to 3×3 Grid. This suggests that the observed affordance gains, particularly in edge alignment, are achieved without increasing geometric deformation relative to prior methods that consider only a single spatial affordance, thereby avoiding additional transformation costs that could undermine motion consistency in heterogeneous spatial layouts.

Finally, the method yields deterministic correspondences: once the mapping is selected before the shared view appears, the same piecewise transformation is applied consistently throughout the session. The $O! \rightarrow P! \rightarrow R$ prioritization, four-type patch taxonomy, and objective-driven selection together provide a geometry- and semantics-aware pathway that preserves task-relevant regions first, avoiding mid-session drift or prediction-based resets. This determinism and affordance-aware placement make the approach suitable for both VR and AR pipelines (e.g., bidirectional, position-based mapping and session-to-session reproducibility), offering a practical foundation for mixed-reality collaboration where stable alignment of walkable areas and object edges is essential.

5.2 Limitations

Despite the strengths of our approach, several limitations remain. First, our method relies on pre-labeled, static scene data for spatial affordance detection and mapping, which restricts its applicability in real-world scenarios where environments are dynamic and affordances may change over time. For practical deployment, it will be necessary to extend the system to sense spatial affordances directly through MR HMDs and adaptively update the mapping in response to environmental changes. Second, the current implementation primarily addresses horizontal surface mapping and does not yet consider vertical variations in elevation, such as stairs, slopes, or multi-level structures; supporting vertical correspondences would enhance spatial completeness and usability. Third, although our method demonstrates strong quantitative performance, further user studies are required to validate its effectiveness in realistic collaborative settings, particularly with respect to user experience, spatial presence, and interaction naturalness. These limitations highlight meaningful directions for future research toward a more adaptive, comprehensive, and user-centered spatial mapping framework.

6 CONCLUSION AND FUTURE WORK

We propose a spatial affordance-aware affine patch mapping for piecewise space transformation in MR collaboration. Our method addresses dissimilarity between two object-rich spaces by composing affine patches that respect both geometric and semantic structure, yielding deterministic correspondences without mid-session changes and avoiding overlaps or discontinuities. Compared with two grid-based base-

lines on heterogeneous space pairs, our approach attains near-ceiling walkable connectivity comparable to 3×3 Grid (while clearly exceeding 1×1 Grid), improves object–edge alignment over 3×3 Grid, and delivers object–plane coverage comparable to 3×3 Grid. Transform magnitude showed no significant differences across methods, indicating cost-neutral deformation. Taken together, these results support our claim that the proposed mapping secures continuous walkable connectivity and finer object-level correspondences across heterogeneous spaces without additional deformation cost, offering a practical and reproducible primitive for AR/VR pipelines where stable, affordance-relevant alignment is critical.

The primary goal of our method is to enable diverse, complex, and realistic interactions among remote users as if sharing the same physical space. To broaden interactive possibilities, we will target real-world rooms with varied furniture shapes, types, sizes, and arrangements, and incorporate contextual factors beyond spatial attributes—such as collaboration goals, participant numbers/roles, and task demands—so that the objective can adapt to collaboration intent. We also plan to validate perceived continuity and task performance through MR telepresence user studies, and to extend the formulation to multi-cluster/multi-room settings and time-varying layouts. Future work will integrate this improved method into MR systems that manage semantic and geometric scene information through automated or user-curated pipelines, enabling practical deployment in real-world spaces and accommodating common furniture relocation scenarios through appropriate scene updates. We expect the proposed approach to serve as a foundational mechanism that adaptively integrates physical environments with collaborative context, thereby enhancing telepresence and supporting broad adoption in MR collaborative applications.

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REFERENCES

- [1] I. Armeni, O. Sener, A. R. Zamir, H. Jiang, I. Brilakis, M. Fischer, and S. Savarese. 3d semantic parsing of large-scale indoor spaces. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1534–1543, 2016. 2
- [2] Y. Cheng, Y. Yan, X. Yi, Y. Shi, and D. Lindlbauer. Semanticadapt: Optimization-based adaptation of mixed reality layouts leveraging virtual-physical semantic connections. In *The 34th Annual ACM Symposium on User Interface Software and Technology*, pp. 282–297, 2021. 2
- [3] B. J. Congdon, T. Wang, and A. Steed. Merging environments for shared spaces in mixed reality. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology*, pp. 1–8, 2018. 2, 3
- [4] T. De Bauw, R. Cools, and A. L. Simeone. Adaptable: Using conformal mapping for collaboration on tables in asymmetric mixed reality. In *Proceedings of the 2022 ACM Symposium on Spatial User Interaction*, pp. 1–2, 2022. 2
- [5] L. Fan, H. Li, and M. Shi. Redirected walking for exploring immersive virtual spaces with hmd: a comprehensive review and recent advances. *IEEE Transactions on Visualization and Computer Graphics*, 2022. 3
- [6] A. Fender, D. Lindlbauer, P. Herholz, M. Alexa, and J. Müller. Heatspace: Automatic placement of displays by empirical analysis of user behavior. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*, pp. 611–621, 2017. 2
- [7] C. G. Fidalgo, M. Sousa, D. Mendes, R. K. Dos Anjos, D. Medeiros, K. Singh, and J. Jorge. Magic: Manipulating avatars and gestures to improve remote collaboration. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 438–448. IEEE, 2023. 3
- [8] J. E. S. Grønbaek, K. Pfeuffer, E. Velloso, M. Astrup, M. I. S. Pedersen, M. Kjær, G. Leiva, and H. Gellersen. Partially blended realities: Aligning dissimilar spaces for distributed mixed reality meetings. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pp. 1–16, 2023. 1, 2
- [9] J. E. S. Grønbaek, J. Sánchez Esquivel, G. Leiva, E. Velloso, H. Gellersen, and K. Pfeuffer. Blended whiteboard: Physicality and reconfigurability in remote mixed reality collaboration. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pp. 1–16, 2024. 1, 2
- [10] C. R. Harris, K. J. Millman, S. J. Van Der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, et al. Array programming with numpy. *Nature*, 585(7825):357–362, 2020. 5
- [11] J. Herskovitz, Y. F. Cheng, A. Guo, A. P. Sample, and M. Nebeling. Xspace: An augmented reality toolkit for enabling spatially-aware distributed collaboration. *Proceedings of the ACM on Human-Computer Interaction*, 6(ISS):277–302, 2022. 2
- [12] H. G. Hoffman. Physically touching virtual objects using tactile augmentation enhances the realism of virtual environments. In *Proceedings. IEEE 1998 Virtual Reality Annual International Symposium (Cat. No. 98CB36180)*, pp. 59–63. IEEE, 1998. 2
- [13] N. Hughes, Y. Chang, and L. Carbone. Hydra: A real-time spatial perception engine for 3d scene graph construction and optimization. *arXiv preprint arXiv:2201.13360*, 2022. 2
- [14] J. D. Hunter. *Matplotlib: A 2D Graphics Environment*, 2007. 5
- [15] J. Kang, D. Yang, T. Kim, Y. Lee, and S.-H. Lee. Real-time retargeting of deictic motion to virtual avatars for augmented reality telepresence. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 885–893. IEEE, 2023. 3
- [16] A. Kendon. *Conducting interaction: Patterns of behavior in focused encounters*, vol. 7. CUP Archive, 1990. 3
- [17] M. Keshavarzi, A. Y. Yang, W. Ko, and L. Caldas. Optimization and manipulation of contextual mutual spaces for multi-user virtual and augmented reality interaction. In *2020 IEEE conference on virtual reality and 3D user interfaces (VR)*, pp. 353–362. IEEE, 2020. 1, 2
- [18] M. Keshavarzi, M. Zollhoefer, A. Y. Yang, P. Peluse, and L. Caldas. Mutual scene synthesis for mixed reality telepresence. *arXiv preprint arXiv:2204.00161*, 2022. 2
- [19] D. Kim, H.-I. Kim, and W. Woo. Mutual space generation with relative translation gains in redirected walking for asymmetric remote collaboration. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 674–677. IEEE, 2022. 3
- [20] D. Kim, S. Kim, S. Choi, and W. Woo. Spatial affordance-aware interactable subspace allocation for mixed reality telepresence. *arXiv preprint arXiv:2408.04297*, 2024. 1, 2, 4
- [21] D. Kim, S. Kim, J.-e. Shin, B. Yoon, J. Kim, J. Lee, and W. Woo. The effects of spatial configuration on relative translation gain thresholds in redirected walking. *Virtual Reality*, pp. 1–18, 2022. 3
- [22] D. Kim, J.-e. Shin, J. Lee, and W. Woo. Adjusting relative translation gains according to space size in redirected walking for mixed reality mutual space generation. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 653–660. IEEE, 2021. 3, 5
- [23] D. Kim and W. Woo. Edge-centric space rescaling with redirected walking for dissimilar physical-virtual space registration. *arXiv preprint arXiv:2308.11210*, 2023. 1, 2, 3, 4, 5, 6
- [24] S. Kim, D. Kim, J.-E. Shin, and W. Woo. Object cluster registration of dissimilar rooms using geometric spatial affordance graph to generate shared virtual spaces. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 796–805. IEEE, 2024. 1, 2, 4, 6
- [25] N. H. Lehment, D. Merget, and G. Rigoll. Creating automatically aligned consensus realities for ar videoconferencing. In *2014 IEEE international symposium on mixed and augmented reality (ISMAR)*, pp. 201–206. IEEE, 2014. 1, 2
- [26] C. Li, W. Li, H. Huang, and L.-F. Yu. Interactive augmented reality storytelling guided by scene semantics. *ACM Transactions on Graphics (TOG)*, 41(4):1–15, 2022. 2
- [27] S. Looper, J. Rodríguez-Puigvert, R. Siegwart, C. Cadena, and L. Schmid. 3d vs: Long-term semantic scene change prediction through 3d variable scene graphs. *arXiv preprint arXiv:2209.07896*, 2022. 2
- [28] W. McKinney et al. Data structures for statistical computing in python. *SciPy*, 445(1):51–56, 2010. 5
- [29] P. Merrell, E. Schkufza, Z. Li, M. Agrawala, and V. Koltun. Interactive furniture layout using interior design guidelines. *ACM transactions on graphics (TOG)*, 30(4):1–10, 2011. 2
- [30] N. Numan, S. Rajaram, B. T. Kumaravel, N. Marquardt, and A. D. Wil-

- son. Spaceblender: Creating context-rich collaborative spaces through generative 3d scene blending. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–25, 2024. 2
- [31] S. Orts-Escolano, C. Rhemann, S. Fanello, W. Chang, A. Kowdle, Y. Degtyarev, D. Kim, P. L. Davidson, S. Khamis, M. Dou, et al. Holoportation: Virtual 3d teleportation in real-time. In *Proceedings of the 29th annual symposium on user interface software and technology*, pp. 741–754, 2016. 1
- [32] M. Pabst, L. Rudolph, N. Brasch, V. Biener, C. Egtebas, U. Eck, D. Schmalstieg, and G. Klinker. Mrunion: Asymmetric task-aware 3d mutual scene generation of dissimilar spaces for mixed reality telepresence. *IEEE Transactions on Visualization and Computer Graphics*, 2025. 2
- [33] P. Panda, L. Tankelevitch, B. Spittle, K. Inkpen, J. Tang, S. Junuzovic, Q. Qi, P. Sweeney, A. D. Wilson, W. A. Buxton, et al. Hybridge: Bridging spatiality for inclusive and equitable hybrid meetings. *Proceedings of the ACM on Human-Computer Interaction*, 8(CSCW2):1–39, 2024. 4
- [34] T. Pejsa, J. Kantor, H. Benko, E. Ofek, and A. Wilson. Room2room: Enabling life-size telepresence in a projected augmented reality environment. In *Proceedings of the 19th ACM conference on computer-supported cooperative work & social computing*, pp. 1716–1725, 2016. 2
- [35] S. Qi, Y. Zhu, S. Huang, C. Jiang, and S.-C. Zhu. Human-centric indoor scene synthesis using stochastic grammar. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 5899–5908, 2018. 2
- [36] X. Qian, F. He, X. Hu, T. Wang, A. Ipsita, and K. Ramani. Scalar: Authoring semantically adaptive augmented reality experiences in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pp. 1–18, 2022. 2
- [37] S. Razzaque, Z. Kohn, and M. C. Whitton. *Redirected walking*. Citeseer, 2005. 3
- [38] P. Schmitz, J. Hildebrandt, A. C. Valdez, L. Kobbelt, and M. Ziefle. You spin my head right round: Threshold of limited immersion for rotation gains in redirected walking. *IEEE transactions on visualization and computer graphics*, 24(4):1623–1632, 2018. 3
- [39] J.-e. Shin, J. Park, H. Kim, W. Woo, and B. Yoon. Evaluating metaverse platforms: Status, direction, and challenges. *IEEE Consumer Electronics Magazine*, 2024. 1
- [40] L. Sidenmark, T. Zhang, L. Al Lababidi, J. Li, and T. Grossman. Desk2desk: Optimization-based mixed reality workspace integration for remote side-by-side collaboration. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–15, 2024. 2
- [41] S. Song, F. Yu, A. Zeng, A. X. Chang, M. Savva, and T. Funkhouser. Semantic scene completion from a single depth image. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1746–1754, 2017. 2
- [42] B. Spittle, P. Panda, L. Tankelevitch, K. Inkpen, J. Tang, S. Junuzovic, Q. Qi, P. Sweeney, A. D. Wilson, W. A. Buxton, et al. Comparing the agency of hybrid meeting remote users in 2d and 3d interfaces of the hybridge system. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pp. 1–12, 2024. 4
- [43] M. Sra, A. Mottelson, and P. Maes. Your place and mine: Designing a shared vr experience for remotely located users. In *Proceedings of the 2018 Designing Interactive Systems Conference*, pp. 85–97, 2018. 1, 2
- [44] F. Steinicke, G. Bruder, J. Jerald, H. Frenz, and M. Lappe. Estimation of detection thresholds for redirected walking techniques. *IEEE transactions on visualization and computer graphics*, 16(1):17–27, 2009. 3
- [45] J. Wald, H. Dhano, N. Navab, and F. Tombari. Learning 3d semantic scene graphs from 3d indoor reconstructions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3961–3970, 2020. 2
- [46] X. Wang, H. Ye, C. Sandor, W. Zhang, and H. Fu. Predict-and-drive: Avatar motion adaption in room-scale augmented reality telepresence with heterogeneous spaces. *IEEE Transactions on Visualization and Computer Graphics*, 28(11):3705–3714, 2022. 3
- [47] N. L. Williams, A. Bera, and D. Manocha. Arc: Alignment-based redirection controller for redirected walking in complex environments. *IEEE Transactions on Visualization and Computer Graphics*, 27(5):2535–2544, 2021. 3
- [48] N. L. Williams, A. Bera, and D. Manocha. Eni: Quantifying environment compatibility for natural walking in virtual reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 419–427. IEEE, 2022. 3
- [49] E. Wong, A. Genay, J. E. Grønbaek, and E. Velloso. Spatial heterogeneity in distributed mixed reality collaboration. In *CHI'25: CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, 2025. 3
- [50] M. Wu, Y. Lu, S. Trivedi, B. Chen, Q. Zhou, L. Wang, S. Singh, M. Zink, R. Sitarman, J. Chakareski, et al. Interactive scene graph analysis for future intelligent teleconferencing systems. In *2023 IEEE International Symposium on Multimedia (ISM)*, pp. 251–255. IEEE, 2023. 2
- [51] S.-Z. Xu, J.-H. Liu, M. Wang, F.-L. Zhang, and S.-H. Zhang. Multi-user redirected walking in separate physical spaces for online vr scenarios. *IEEE Transactions on Visualization and Computer Graphics*, 2023. 2
- [52] B. Yoon, H.-i. Kim, G. A. Lee, M. Billingham, and W. Woo. The effect of avatar appearance on social presence in an augmented reality remote collaboration. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 547–556. IEEE, 2019. 1
- [53] L. Yoon, D. Yang, C. Chung, and S.-H. Lee. A full body avatar-based telepresence system for dissimilar spaces. *arXiv preprint arXiv:2103.04380*, 2021. 3
- [54] L. Yoon, D. Yang, J. Kim, C. Chung, and S.-H. Lee. Placement retargeting of virtual avatars to dissimilar indoor environments. *IEEE Transactions on Visualization and Computer Graphics*, 28(3):1619–1633, 2020. 2, 3
- [55] F. Zaman, C. Anslow, A. Chalmers, and T. Rhee. Mrmac: Mixed reality multi-user asymmetric collaboration. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 591–600. IEEE, 2023. 1
- [56] S.-K. Zhang, W.-Y. Xie, and S.-H. Zhang. Geometry-based layout generation with hyper-relations among objects. *Graphical Models*, 116:101104, 2021. 2